

HybridDeform4D: Joint Refinement of Mesh and Gaussian Splatting for Video-to-4D Object Generation

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Abstract

Generating dynamic, spatiotemporally consistent 3D content from monocular video is a significant challenge. Existing methods leveraging Gaussian Splatting (GS) often struggle to maintain such coherence without strong regularization. We introduce *HybridDeform4D*, a novel framework that enhances 4D object generation by jointly optimizing a deformable mesh with its surface-bound GS attributes. This framework supports starting with a coarse mesh and capturing both complex motion and appearance variations without requiring high-quality initialization. To ensure coherent animation, we introduce a coarse-to-fine optimization scheme. This strategy first focuses on learning motion around a key reference frame before progressively expanding to optimize the entire video sequence, ensuring stable and accurate temporal dynamics. Experimental results show that *HybridDeform4D* achieves high rendering quality and spatial-temporal consistency. Furthermore, we demonstrate the versatility of our approach by extending it to text-to-3D synthesis.

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1 Introduction

Recent advancements in 4D synthesis build upon score distillation sampling (SDS) [Poole et al. 2022] from 2D diffusion models to generate dynamic 3D content. While recent methods using GS [Kerbl et al. 2023] representation are quite popular, they often model motion by optimizing each Gaussian independently [Ren et al. 2022; Zeng et al. 2024], creating an unstructured and inefficient space for learning deformations. DreamMesh4D [Li et al. 2024] leverages a deformable mesh to govern the motion of surface-bound Gaussians (i.e., SuGaR [Guédon and Lepetit 2024]). This approach enforces spatial coherence through the mesh’s connectivity. However, DreamMesh4D struggles to capture intricate appearance changes or topological variations like object-part overlaps (Figure 3), which are not easily represented by mesh transformations alone.

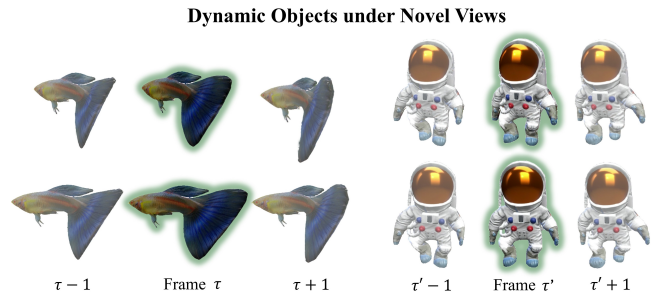


Figure 1: Generated 4D assets through HybridDeform4D.

We introduce *HybridDeform4D*, a coarse-to-fine framework for video-to-4D generation. As illustrated in Figure 2, our method first generates a textured mesh from a key reference frame using image-to-3D techniques [Liu et al. 2023; Xiang et al. 2025]. Then, a dynamic learning stage refines both geometry and appearance through the

joint optimization of the mesh vertices and attributes of bound surface Gaussians. To achieve high-quality mesh deformations, we reparameterize the vertices ($V \rightarrow V^* = (I + \lambda L)$) based on the Laplace-Beltrami operator L [Nicolet et al. 2021].

2 Method

HybridDeform4D starts by attaching 6 Gaussians to each face of an initial mesh, adopting the differentiable rendering pipeline and loss designs from DreamMesh4D. After rendering its RGB $\hat{C}_\tau^*$ and alpha $\hat{S}_\tau^*$ under reference view, we compute reconstruction loss $\mathcal{L}_{\text{ref}} = \|\hat{C}_\tau^* - C_\tau^*\|$ and mask loss $\mathcal{L}_{\text{mask}} = \|\hat{S}_\tau^* - S_\tau^*\|$, where C_τ^* and S_τ^* are the ground-truth data from input video at timestamp τ .

$$\mathcal{L}_{4D} = \lambda_{\text{SDS}} \mathcal{L}_{\text{SDS}} + \lambda_{\text{ref}} \mathcal{L}_{\text{ref}} + \lambda_{\text{mask}} \mathcal{L}_{\text{mask}} + \lambda_{\text{ARAP}} \mathcal{L}_{\text{ARAP}} \quad (1)$$

The hybrid representation is optimized with the above composite weighted loss function, where an SDS loss is utilized for supervision under other randomly sampled views. As-Rigid-As-Possible (ARAP) regularization is incorporated to preserve geometric integrity.

A core component of our method is a Multi-Layer Perceptron (MLP) that explicitly models the joint optimization of time-varying geometry and appearance. Conditioned on the embedding of frame τ , this network predicts the deformed mesh vertex positions (V^*) as well as the dynamic attributes of bound Gaussians (i.e., opacity and Spherical Harmonic (SH) coefficients).

To ensure temporally coherent results, we employ a coarse-to-fine optimization strategy. The training process initially focuses on a narrow window of frames around the reference view (e.g., $\tau_0 \pm 1$). This temporal window then linearly expands throughout the first 70% of the optimization steps until it encompasses the entire video sequence. Frame sampling within this window is hybrid: 70% of frames are chosen randomly to capture diverse states, while 30% are selected sequentially to maintain local motion dynamics.

3 Evaluation

For quantitative analysis, we adopted the metrics from Stag4D and DreamMesh4D. We evaluated image-level fidelity using the CLIP score and reference-view PSNR. Temporal coherence was assessed with the FID-VID. To measure mesh quality during deformation, the self-intersection ratio was calculated. As detailed in Table 1, our framework consistently outperforms baseline approaches.

An ablation study was conducted to validate our joint optimization and coarse-to-fine strategy. As shown in Figure 3, joint optimization is critical for capturing fine details, such as the movement of the elephant’s eye, while the coarse-to-fine approach successfully models large-scale motions, like the frog’s arm lifting. Furthermore, we demonstrate the versatility of our joint refinement process, which can be extended to other tasks like optimization-based image-to-3D generation (Figure 4).

Table 1: Quantitative comparison with baseline methods.

Method	PSNR (ref) $\uparrow$	FVD $\downarrow$	SSIM (ref) $\uparrow$	Self-Intersection $\downarrow$
DreamMesh4D [Li et al. 2024]	36.68	498.47	0.858	0.69%
Stag4D [Zeng et al. 2024]	33.55	610.77	0.844	-
Ours	38.03	427.02	0.865	0.18%

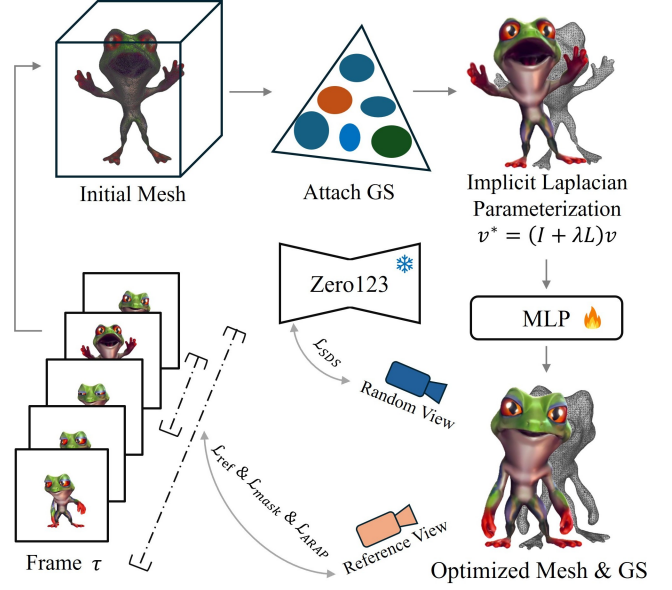


Figure 2: Overview of the HybridDeform4D pipeline. Our pipeline begins by generating a hybrid mesh-Gaussian representation from a single reference frame. We then jointly optimize the mesh deformation and the GS attributes for dynamic appearance.

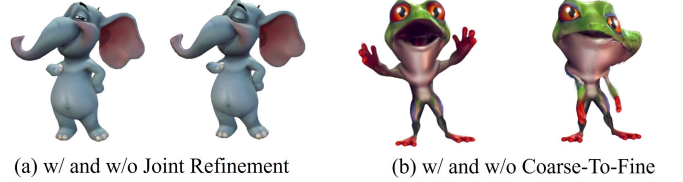


Figure 3: Ablation study results.

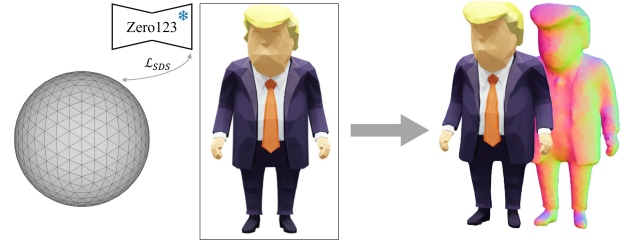


Figure 4: Applying our joint refinement to 3D generation. Mesh deformation serves as an effective geometric prior to improve Gaussian optimizations.

4 Discussion

HybridDeform4D enhances 4D object generation by adopting joint refinement optimizations for both mesh deformation and GS appearance. This strategy effectively supports the spatial-temporal consistency during dynamic generation. Future work can expand our approach to feed-forward training and scene-level generation.

References

- Antoine Guédon and Vincent Lepetit. 2024. SuGaR: Surface-Aligned Gaussian Splatting for Efficient 3D Mesh Reconstruction and High-Quality Mesh Rendering. In *CVPR*.
- Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 2023. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Trans. Graph.* 42, 4 (2023).
- Zhiqi Li, Yiming Chen, and Peidong Liu. 2024. Dreammesh4D: Video-to-4D Generation with Sparse-Controlled Gaussian-Mesh Hybrid Representation. *Advances in Neural Information Processing Systems* 37 (2024), 21377–21400.
- Ruoshi Liu, Rundt Wu, Basile Van Hoorick, Pavel Tokmakov, Sergey Zakharov, and Carl Vondrick. 2023. Zero-1-to-3: Zero-shot One Image to 3D Object. *arXiv preprint arXiv:2303.11328* (2023).
- Baptiste Nicolet, Alec Jacobson, and Wenzel Jakob. 2021. Large Steps in Inverse Rendering of Geometry. *ACM Trans. Graph.* 40, 6, Article 248 (2021), 13 pages.
- Ben Poole, Ajay Jain, Jonathan T. Barron, and Ben Mildenhall. 2022. DreamFusion: Text-to-3D Using 2D Diffusion. *arXiv preprint arXiv:2209.14988* (2022).
- Jiawei Ren, Liang Pan, Jiaxiang Tang, Chi Zhang, Ang Cao, Gang Zeng, and Ziwei Liu. 2022. DreamGaussian4D: Generative 4D Gaussian Splatting. *arXiv preprint arXiv:2312.17142* (2022).
- Jianfeng Xiang, Zelong Lv, Sicheng Xu, Yu Deng, Ruicheng Wang, Bowen Zhang, Dong Chen, Xin Tong, and Jiaolong Yang. 2025. Structured 3D Latents for Scalable and Versatile 3D Generation. In *CVPR*.
- Yifei Zeng, Yanqin Jiang, Siyu Zhu, Yuanxun Lu, Youtian Lin, Hao Zhu, Weiming Hu, Xun Cao, and Yao Yao. 2024. STAG4D: Spatial-Temporal Anchored Generative 4D Gaussians. In *ECCV*. Berlin, Heidelberg, 163–179.